1. Why it feels the way it does (Swink, 2009)

The brief said the controller should not feel floaty, stiff, or slow to respond. I used Swink's (2009) ideas about input and response to get there. For input, I put the controls on different fingers. WASD or the left stick moves, Space jumps, and Ctrl or C crouches. Nothing the player needs to press together fights for the same finger. I also removed walking. On a keyboard, a half-speed "walk" is a hidden mode that the player never asked for. Now any movement input makes the character run, and the blend tree eases her up from standing still.

For response, I thought about the ADSR curve from our lectures. The speed value changes in 0.07 seconds, so she starts moving fast. She turns at 840 degrees per second, so direction changes are quick. The animation's own movement decides how she stops, so she does not slide. The jump rises fast and falls faster, and letting go of the button early makes a shorter jump. I also added two helpers from the Week 3 lecture: coyote time (0.12 s), which lets you jump just after leaving a ledge, and an input buffer (0.15 s), which remembers a jump press made just before landing.

I picked these numbers by trying them. I did not run a formal playtest. The prototype also has no sound. Week 4 (Horowitz & Looney, 2014) shows how much games rely on audio for feedback, so this is a real weakness, and I list it as a limit, not a choice.

2. Core engine systems

- **New Input System:** one actions file with Move, Jump and Crouch for keyboard and gamepad. A small script reads it and remembers jump presses.
- **Animator and Humanoid Avatar:** Eve and all the clips use the Humanoid type, so the idle, walk and run clips from another pack work on her skeleton. A blend tree mixes idle, walk and run using a Speed value. Jump, Fall and Crouch are separate states.
- **CharacterController:** one script (PlayerMotor) moves her using the animation's own movement, so her feet match her speed. It also handles gravity, jumping and the crouch size.
- **Cinemachine:** a FreeLook-style camera with damping and dead zones so it feels like it has weight. Low-strength noise adds a light breathing sway.
- **Particle system:** a dust effect appears at her feet when she moves fast enough.

3. Hardest technical problem

The hardest problem was making the jump animation match the jump physics. The Mixamo jump clip is 1.9 seconds long and starts with a long crouch. But my physics jump (1 m high) keeps her in the air for only about 0.6 seconds. Played from the start, she crouched for the whole jump and showed the leap pose after landing. It looked wrong.

I kept the jump in code, because animation-driven jumps are hard to tune for height. Instead I changed the animation timing. The Jump state starts partway into the clip and plays 1.4 times faster. Then it fades into a Fall state that holds the in-air pose until she touches the ground. So the animation follows the physics, not the other way around.

4. Theory: enacted narrative (Jenkins, 2004)

Jenkins (2004) says game spaces can tell stories through how the player moves in them, not through words. My grey-box course has no story text, but its order still tells a story about what the character can do. The player starts on a flat area, runs past posts, climbs steps and a ramp, drops off a platform, jumps over blocks that get taller, and finally reaches a low ceiling (1.3 m, while she is 1.35 m tall) that she can only pass by crouching. Each section teaches one new move before the next appears. I kept everything plain grey with no signs on the entrances so the shapes alone show this order. Kremers (2009) also covers how level structure is planned.

5. Where the engine forced a compromise

Unity's CharacterController only tells you if you were on the ground after the last move, and the game runs at a changing frame rate. On slopes and step edges, the "on the ground" value would flicker for one frame, and the animation briefly switched to Fall. I fixed this in two ways. A small constant downward push keeps her on the floor, and the animator also checks the ground with a short downward probe.

Crouching had a similar problem. A character cannot shrink and grow smoothly inside solid objects. So her capsule shrinks to 0.85 m, and she will not stand up while something is above her head. If you let go of the crouch key under the low ceiling, she stays crouched. In both cases I turned a smooth idea into a simple rule that is checked every frame.

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Assets

https://www.mixamo.com/#/?page=1&query=eve&type=Character